

Muhammad Sayyedain

Software Engineer • Python Backend • Full-Stack Development

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SUMMARY

BSCS graduate with hands-on experience building backend APIs and full-stack applications using Python, FastAPI, Flutter, and Firebase. Comfortable with object-oriented programming, data structures, databases, REST APIs, Git, and debugging application issues. Also experienced in building retrieval and LLM-based systems.

TECHNICAL SKILLS

Languages: Python, Dart, SQL

Backend: FastAPI, REST APIs, Pydantic, PostgreSQL, Firebase, Supabase

Computer Science: Object-Oriented Programming, Data Structures and Algorithms, DBMS, Software Engineering

AI and Retrieval: LangChain, LangGraph, RAG, ChromaDB, FAISS, BM25, RRF, cross-encoder reranking

Mobile and Frontend: Flutter, Provider, Firebase Authentication, Firestore

Tools: Git, GitHub, Docker, Linux, Postman, JMeter, Hugging Face

EXPERIENCE

OptimusFox — Lahore, Pakistan

Jun 2025 – Jul 2025

Mobile App Development Intern [\[Code\]](#)

- Developed features for a Flutter and Firebase property-booking application, including property listings, saved properties, booking workflows, and an admin interface.
- Designed Firestore collections and queries for properties, users, favorites, and bookings.
- Worked with reusable Flutter components, application state management, Firebase Authentication, and backend data integration.

PROJECTS

Customs Automation and Compliance Engine [\[Code\]](#)

2026

Python, FastAPI, PostgreSQL, PyMuPDF, Pydantic, Docker

- Building a FastAPI backend for uploading, storing, extracting, and reviewing customs and export documents.
- Implemented PDF validation, UUID-based file storage, document metadata tracking, PostgreSQL persistence, and structured API responses using Pydantic.
- Organized the backend into route, service, schema, dependency, extraction, and validation layers to keep the code easier to test and debug.
- Added PDF text extraction using PyMuPDF as the first stage of a document-compliance pipeline.

Healthcare RAG Assistant [\[Code\]](#)

2026

Python, FastAPI, LangChain, ChromaDB, FAISS, BM25, RAGAS

- Built a clinic assistant that answers patient questions using the clinic's own policy and scheduling documents through a FastAPI service.
- Combined vector retrieval with BM25 keyword search, merged results using reciprocal rank fusion, and reranked them with a cross-encoder.
- Added multi-turn conversation memory and a preprocessing step that removes CNIC and phone-number information before model requests.
- Evaluated retrieval and answer quality using RAGAS and investigated failed queries using LangSmith traces.

PalPath — Dental X-ray Analysis Platform [\[Code\]](#)

2025 – 2026

Flutter, Python, FastAPI, YOLOv8, Firebase, Supabase

- Built a dental case-management application with separate workflows for students, instructors, and practitioners.
- Implemented patient records, case notes, appointments, treatment plans, quizzes, and role-based application screens.
- Connected the Flutter application to a Python API that runs a YOLOv8 model on uploaded dental X-rays and returns detected findings.

EDUCATION

Bahria University — Lahore, Pakistan

Sep 2022 – Jun 2026

Bachelor of Science in Computer Science

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Systems, Software Engineering

CERTIFICATIONS AND TRAINING

Crewlogix PSDF Generative AI Bootcamp

May 2026 – Jul 2026

13-week applied training covering RAG systems, LLM agents, evaluation, fine-tuning, FastAPI deployment, and Docker.

Supervised Machine Learning: Regression and Classification — Stanford Online, Coursera | [Certificate](#)